

The Richness Spectrum of Quadratic Primes on the $6N$ Skeleton: Mask Rotation in $n^2 + a$ and the Heegner Corridor of $n^2 + n + A^*$

Ruqing Chen

GUT Geoservice Inc., Montreal, Quebec, Canada

ruqing@hotmail.com

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Abstract

Part XXIX–XXXI mapped the linear constellations and the single nonlinear sequence $n^2 + 1$. Here we vary the quadratic and read the whole family through one principle: a quadratic generates primes densely exactly when its discriminant *dodges* the low primes—when no small q can divide its values. We measure two families by a square-root sieve. (i) For $n^2 + a$, the shift a rotates a geometric mask: $a \bmod 6$ decides which break-zones are open. We find $a \equiv 1, 4$ feed both wings at 1:2; $a \equiv 0, 3$ seal the left zone and feed only the right; $a \equiv 2, 5$ seal the right and feed only the left. Independently of the mask, the richness $C(a) = \prod_q (1 - \omega_a(q)/q)/(1 - 1/q)$ swings widely with a —from 0.53 ($a = 5$) to 1.97 ($a = 7$)—and the measured count $Q_a(N)/\int dt/\log(t^2+a)$ tracks it across the family. (ii) For Euler’s $n^2 + n + A$, the discriminant is $1 - 4A$, and the richness is governed by how long a *corridor* of small primes the polynomial threads without ever being divisible. The class-number-one (Heegner) values $A = 3, 5, 11, 17, 41$, with $4A - 1 = 11, 19, 43, 67, 163$, give the longest corridors: the first prime that can divide $n^2 + n + A$ is A itself, and the richness climbs monotonically 1.02, 1.88, 3.26, 4.17, 6.64. For $n^2 + n + 41$ we confirm $\omega(q) = 0$ for *every* prime $q \leq 37$ —it dodges all of them—while a non-Heegner neighbour is divisible already at $q = 3$. The two families are one phenomenon: prime richness is the product of per-prime dodges, and the class-number-one discriminants are the champion dodgers, with -163 the largest. None of this is new as theory—Rabinowitsch, Heegner–Stark–Baker, and Bateman–Horn settled it—and we make no claim about the infinitude of any prime-rich polynomial. As throughout, this is a measurement, not a theorem.

1 Introduction

With $n^2 + 1$ (Part XXXI [8]) the programme left the linear world—twin shifts, Goldbach sums, the arithmetic progressions of Part XXX [7]—for its first quadratic. A single sequence, though, only hints at the structure. The richness of $n^2 + 1$ came from a constant $C = 1.37281$ built from local factors that split the primes by the character of -1 . Change the quadratic and that constant moves. This paper varies it deliberately, across two families, and finds that one idea organises both: a quadratic is prime-rich exactly insofar as its discriminant is a non-residue modulo the small

*Code and measured data: <https://github.com/Ruqing1963/6N-quadratic-spectrum>

$a \bmod 6$	open wings	active $n \bmod 6$	example richness $C(a)$
1	both, right:left = 1:2	0 (R), 2, 4 (L)	$a=1: 1.37, a=7: 1.97, a=13: 1.29$
4	both, right:left = 1:2	3 (R), 1, 5 (L)	$a=4: 1.37, a=-2: 1.85$
3	right only (left sealed)	2, 4 (R)	$a=3: 1.12$
0	right only (left sealed)	1, 5 (R)	$a=6: 0.71$
2	left only (right sealed)	3 (L)	$a=2: 0.71$
5	left only (right sealed)	0 (L)	$a=5: 0.53, a=11: 0.51$

Table 1: The mod-6 mask for $n^2 + a$. The shift seals or opens the break-zones; $a \equiv 1, 4$ reproduce the 1:2 wing split of $n^2 + 1$, while $a \equiv 2, 5$ and $a \equiv 0, 3$ extinguish one wing entirely.

primes, so that those primes are *forbidden* to divide its values and act as enhancements rather than sieves.

That a polynomial can dodge the low primes is the old observation behind Euler’s $n^2 + n + 41$, prime for $n = 0, \dots, 39$. Rabinowitsch [2] tied this to the class number, and the Heegner–Stark–Baker theorem [3, 4, 5] fixed the complete list of imaginary quadratic discriminants of class number one, the largest being -163 . The conjectural counts are the Bateman–Horn specialisation of Hardy–Littlewood [1, 6]. None of this is ours. What we add is the $6N$ reading: the shift a as a rotation of the break-zone mask, the discriminant as the setting of a dodge-corridor, and a clean high-precision measurement of the resulting richness spectrum via a square-root sieve. We touch no infinitude question.

2 Setup: discriminant, dodging, and the local factor

For a quadratic f and a prime q , let $\omega(q)$ be the number of roots of $f \equiv 0 \pmod{q}$. A prime can divide the values of f only when $\omega(q) > 0$, which for $q \nmid \text{disc}(f)$ happens exactly when $\text{disc}(f)$ is a quadratic residue mod q . The Bateman–Horn richness is the product of local factors

$$C = \prod_q g(q), \quad g(q) = \frac{1 - \omega(q)/q}{1 - 1/q}, \quad (1)$$

and $Q(N) := \#\{n \leq N : f(n) \text{ prime}\} \sim C \int_2^N dt / \log f(t)$. The factor $g(q)$ is the whole story: where f *cannot* be divided ($\omega = 0$) it reads $q/(q-1) > 1$, an *enhancement*, a dodged prime; where f is divisible on two residues it reads $(q-2)/(q-1) < 1$, a *suppression*. Richness is a tug-of-war between dodged and hit primes, and a discriminant that is a non-residue mod many small q wins it. We enumerate by a square-root sieve on n : for each q with $\omega(q) > 0$ we solve $f(n) \equiv 0$ and strike the offending residues, leaving the primes; here $n^2 + a$ uses $n \equiv \pm\sqrt{-a}$, and $n^2 + n + A$ uses $(2n+1)^2 \equiv 1 - 4A$.

3 The shift a rotates the mask: $n^2 + a$

Modulo 6 the squares are $n^2 \in \{0, 1, 3, 4\}$ (at $n \equiv 0; \pm 1; 3; \pm 2$), so $n^2 + a$ lands in a break-zone ($\equiv \pm 1 \pmod{6}$) only for some residues of n , and which ones depends on $a \bmod 6$. The result (Table 1) is a rotating mask:

The mask sets *which* zones are populated; the richness $C(a)$ of (1) sets *how densely*, and the two are independent. Figure 1(A) shows $C(a)$ over $a = -3, \dots, 48$, coloured by mask: the same green “both-wing” mask carries values from 1.0 to 2.3, and the sealed-wing masks span 0.5 to 1.2.

A	disc $1 - 4A$	class no. 1	first prime $ f$	measured C
3	-11	yes	3	1.018
5	-19	yes	5	1.883
11	-43	yes	11	3.262
17	-67	yes	17	4.174
41	-163	yes	41	6.637
9	-35	no	3	0.927
21	-83	no	3	0.970
33	-131	no	3	0.659

Table 2: The class-number-one ladder for $n^2 + n + A$. For the Heegner values the first divisor is A itself and the richness rises to 6.64 at $n^2 + n + 41$; non-Heegner values are divisible at 3 and stay poor.

The richness is a genuine arithmetic spectrum riding on top of the geometric mask. It is no fit: feeding the local product (1) into Bateman–Horn predicts the measured counts across the whole family to a fraction of a percent (Fig. 1(B); $a = 7$ gives measured 1.975 against predicted 1.973, $a = 5$ gives 0.531 against 0.528). The degenerate $a = -1$, where $n^2 - 1 = (n - 1)(n + 1)$ factors, has $C \rightarrow 0$, the product correctly foreseeing the algebraic collapse.

4 The corridor of $n^2 + n + A$ and the Heegner ladder

Euler’s quadratic has discriminant $1 - 4A$, and its values never leave the break-zones: $n(n+1) + A \equiv \pm 1 \pmod{6}$ for all n when A is odd, so 2 and 3 never divide it. The deeper fact is how far this dodging runs. For $A = 41$ the discriminant is -163 , and we confirm by direct root count that $\omega(q) = 0$ for *every* prime $q \leq 37$: the polynomial threads a corridor past all eleven of them and is first divisible at $q = 41$ (Fig. 2(A), shaded). A non-Heegner neighbour such as $n^2 + n + 9$ (discriminant -35) is already divisible at $q = 3$.

This is the class-number-one phenomenon. Rabinowitsch’s criterion makes $n^2 + n + A$ prime for $n = 0, \dots, A - 2$ precisely when $1 - 4A$ has class number one; the Heegner–Stark–Baker list then forces $A \in \{3, 5, 11, 17, 41\}$ (with $4A - 1 = 11, 19, 43, 67, 163$). For each, the first prime that can divide is A itself— $f(A - 1) = A^2$ —so the corridor length grows with A , and with it the richness (Table 2, Fig. 2(B)): the measured C climbs monotonically 1.02, 1.88, 3.26, 4.17, 6.64, topping out at $n^2 + n + 41$. The non-Heegner values sit well below, and $n^2 + n + 7$ (discriminant $-27 = 9 \cdot (-3)$) lands exactly on $n^2 + n + 1$ (discriminant -3), as it must, the two sharing a character class.

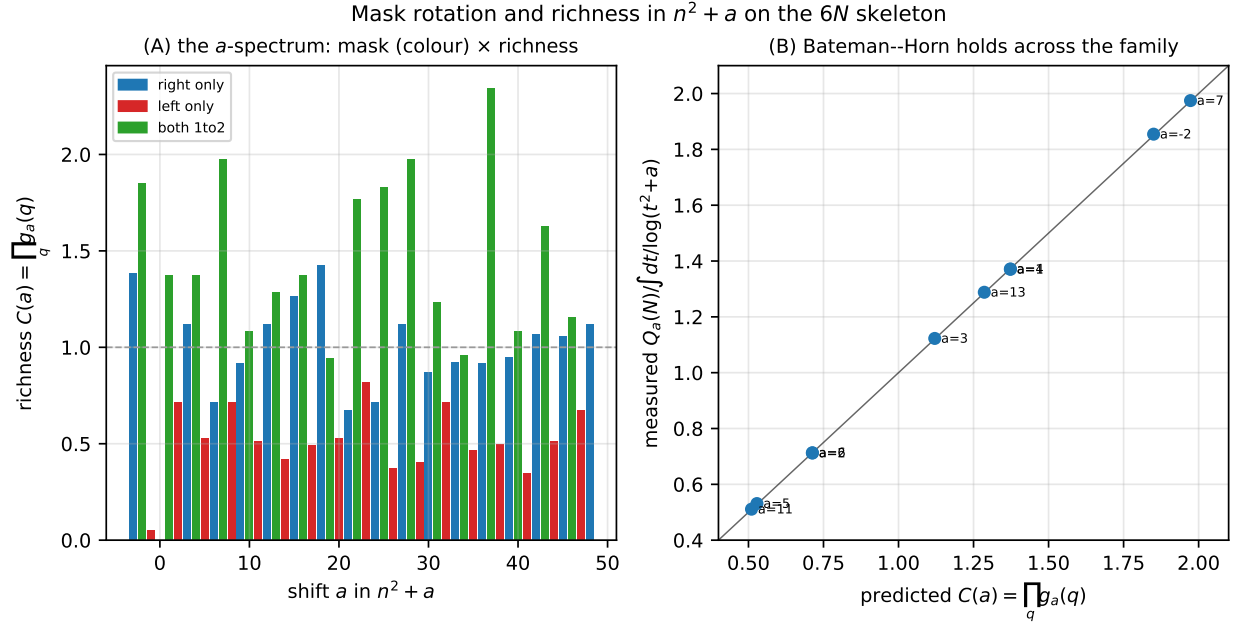


Figure 1: (A) The a -spectrum of $n^2 + a$: richness $C(a)$ versus shift a , coloured by the mod-6 mask (both-wing, right-only, left-only). The geometric mask and the arithmetic richness are independent. (B) The local product (1) against the measured count for representative a , on the diagonal: Bateman–Horn holds across the family.

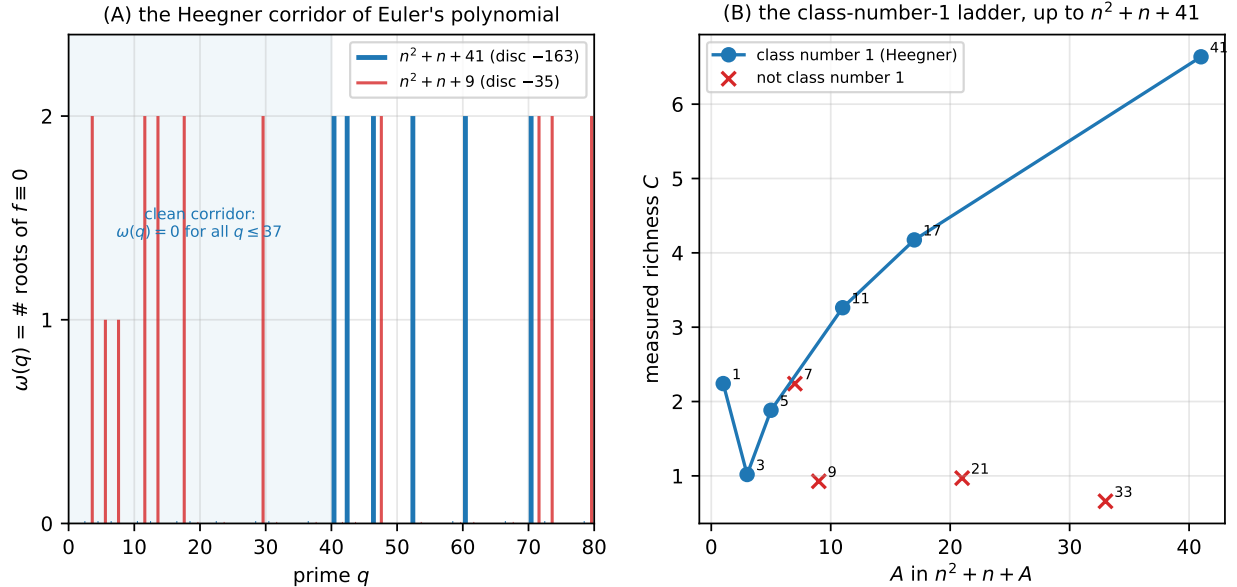


Figure 2: (A) The Heegner corridor: $\omega(q)$ for $n^2 + n + 41$ (disc -163) is zero for every prime $q \leq 37$ (shaded), first nonzero at $q = 41$, while $n^2 + n + 9$ (disc -35) is divisible already at small q . (B) The richness C versus A : the class-number-one values (Heegner) climb to 6.64 at $n^2 + n + 41$; non-Heegner values sit below.

5 Scope and honest ledger

Two families, one principle. The shift a in $n^2 + a$ rotates a mod-6 mask that opens or seals the break-zones— $a \equiv 1, 4$ feeding both at 1:2, $a \equiv 2, 5$ and $a \equiv 0, 3$ extinguishing a wing—while the richness $C(a)$ rides independently on top, swinging from 0.53 to nearly 2 as the discriminant $-4a$ dodges more or fewer small primes. The shift A in $n^2 + n + A$ sets the length of a clean corridor: the class-number-one discriminants thread the most primes, first divisible at A itself, and the richness climbs monotonically to 6.64 at the Heegner champion -163 . Both are the same statement about the local factor $g(q)$: prime richness is the product of dodged primes against hit primes.

We are explicit about the boundary. The mask is elementary; the local factors, the constant C , and the Bateman–Horn asymptotic are classical; the class-number-one criterion is Rabinowitsch’s and the finiteness of the Heegner list is the Heegner–Stark–Baker theorem. What this paper contributes is the $6N$ geometric reading—mask rotation and wing sealing, the corridor as a run of dodged annihilation channels—and a high-precision measurement of the richness spectrum across both families via the square-root sieve. We make no claim about the infinitude of primes of any of these forms; that lies beyond every finite computation, and the counts we report would look the same whatever the truth. Everything here equals the discrete sieve count, takes the Hardy–Littlewood heuristic as input, and makes no claim about the infinitude of any sequence.

Data and code availability. The square-root sieves for $n^2 + a$ and $n^2 + n + A$, the richness scan, the Heegner-family counts, the corridor root-counts, and the figure scripts are openly available at <https://github.com/Ruqing1963/6N-quadratic-spectrum>.

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